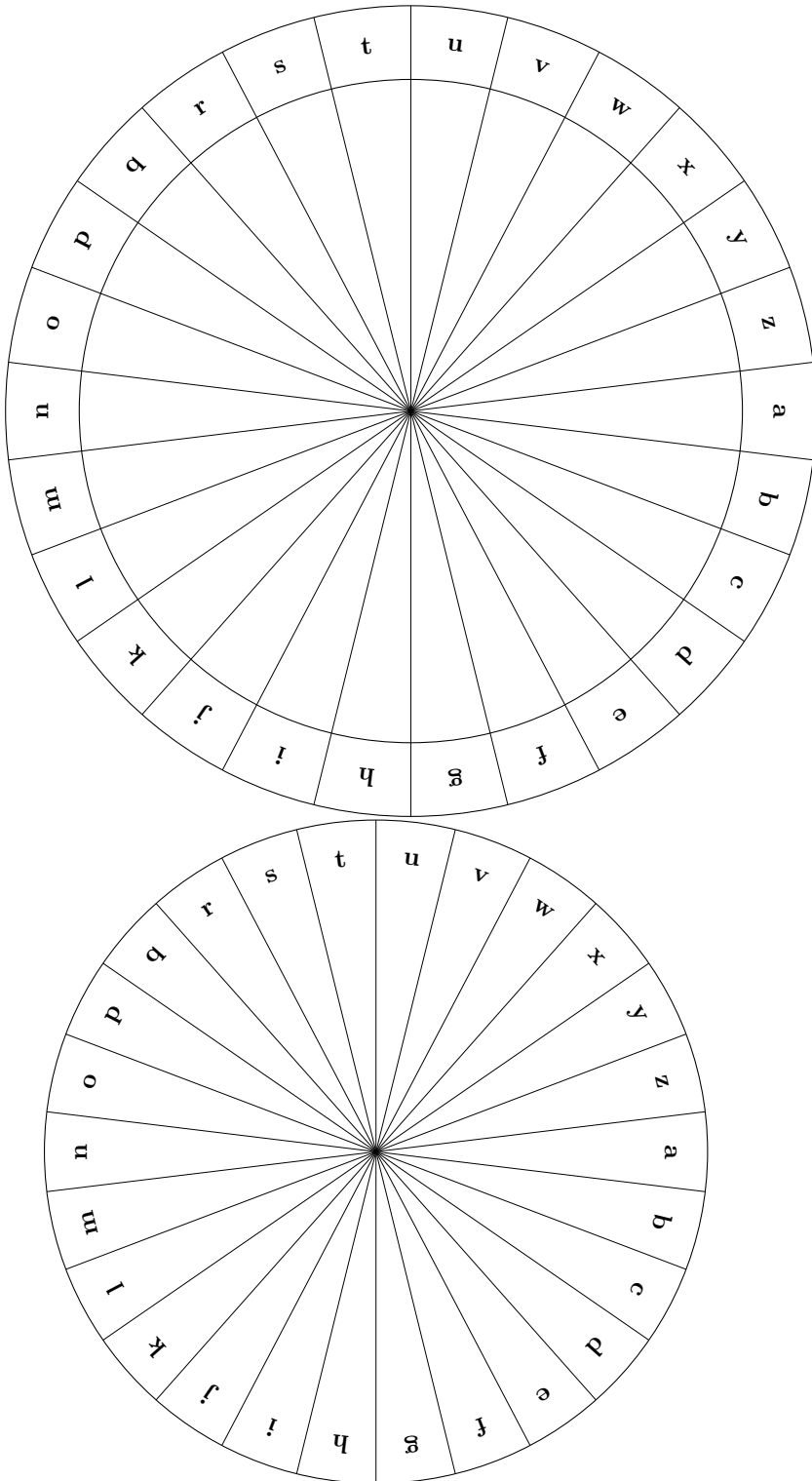


# 1 Materials

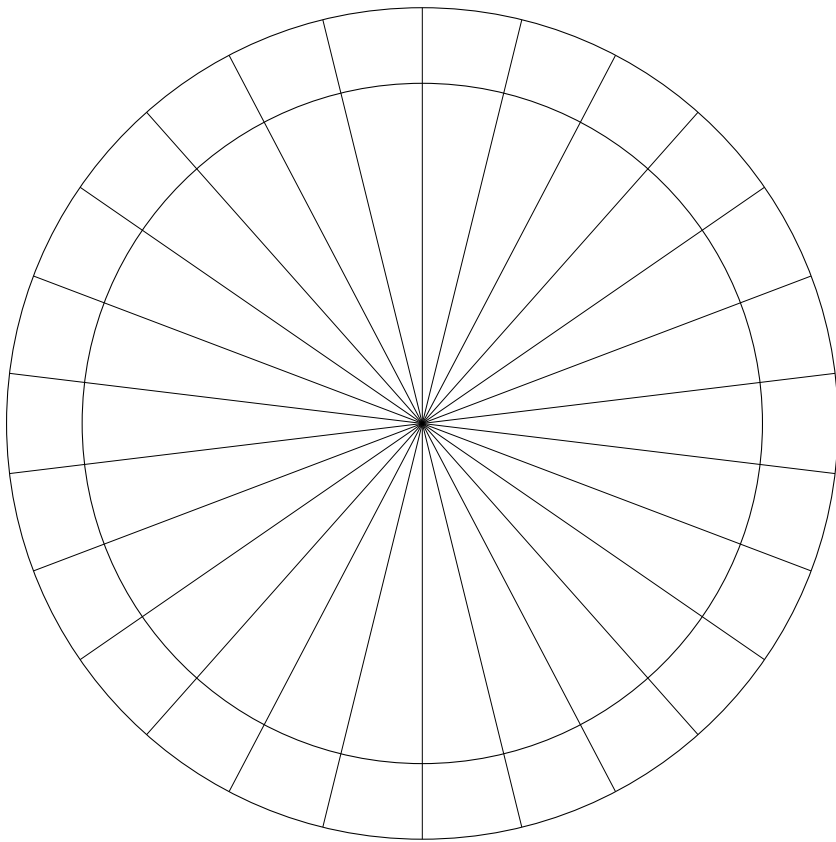
Snap Fastener

## 2 Classical Cryptography exercises

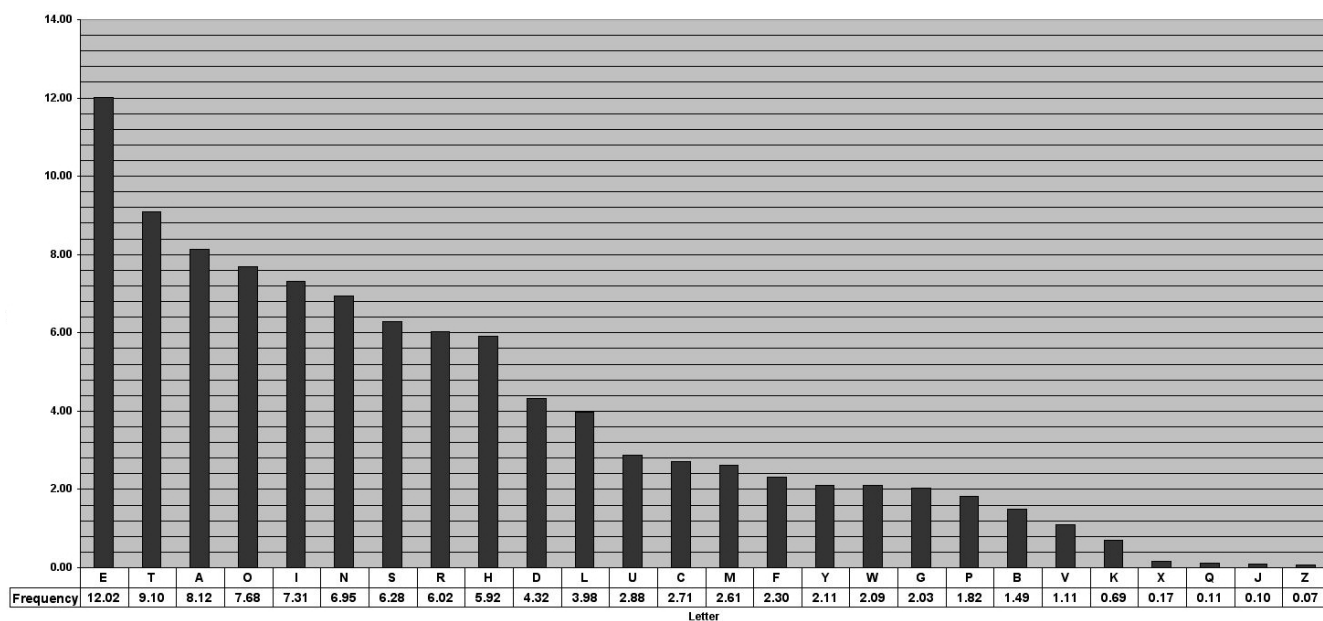
### 2.1 Caesar cipher - A type of substitution cipher



Build your own



1. Hohphqwdub, pb ghdu Zdwvrq
2. Bw mdmzg ikbqvw bpmzm qa iteiga iv mycit ivl wxxwaqbm zmikbqvw
3. Evmvi crlxy rk czmv uirxfej
4. Hdojxcjiymdv dn ocz kjrzmcpnz ja ocz xzgg
1. (3) Elementary, my dear Watson
2. (8) To every action there is always an equal and opposite reaction
3. (17) Never laugh at live dragons.
4. (21) Mitochondria is the powerhouse of the cell



English Letter Frequency (based on a sample of 40,000 words) <https://pi.math.cornell.edu/mec/2003-2004/cryptography/subs/fre>

## 2.2 QR code

## 3 Concept

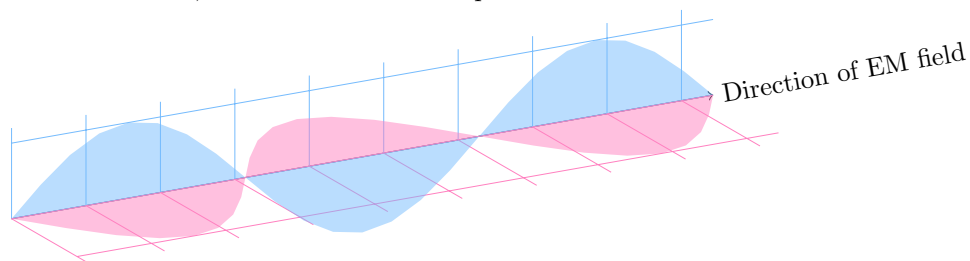
Light is made of photons. It is an em wave with a polarization.

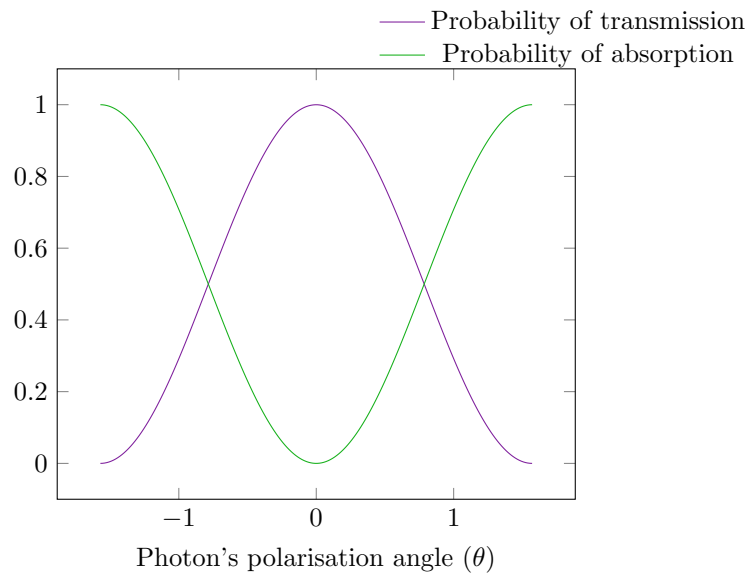
### 3.1 Light as a wave

Has a defined frequency and wavelength, delocalized, interference phenomenon

### 3.2 Light as a particle

Photoelectric effect, made of chunks called photons.

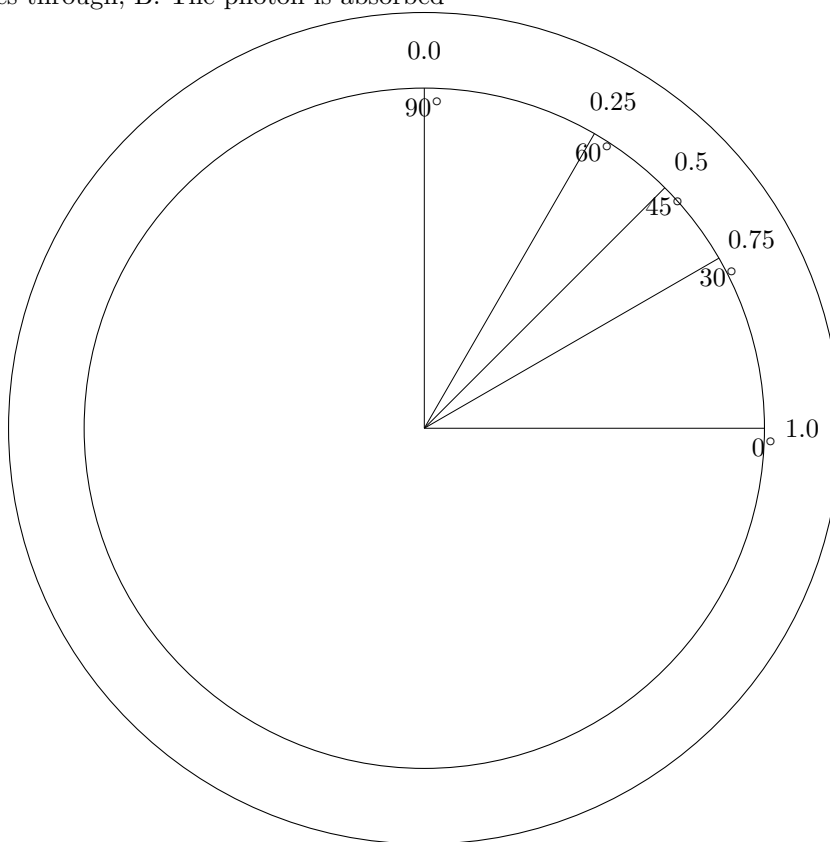




The direction of the electric field is called the light's polarization.

Polarizers are special sheets of material that absorb or transmit light depending on its polarization.

Consider a single photon at a polarizer. There are only two possible things that can happen: A. The photon goes through; B. The photon is absorbed



Mutually exclusive

Vertical and Horizontal are mutually exclusive

vertical and diagonal (45°) polarizations are not mutually exclusive

45° and -45° polarizations are mutually exclusive

Any two perpendicular polarizations are mutually exclusive.

### 3.3 Superposition

A quantum object can be in a superposition of multiple states at once. For example, a  $+45^\circ$ -polarized photon is in a superposition of horizontal and vertical polarization states.

### 3.4 Measurement

Superposition works as long as you don't look! The act of measuring the superposition will collapse it and change the state. For example, when we measure the  $+45^\circ$ -polarized photon in the HV basis, it must collapse to either horizontal or vertical polarization

